

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EE-603 (GS)
B.E. VI Semester Examination, June 2020
Grading System (GS)
Electrical Machine Design
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Find the suitable number of poles and the dia of the core of a 400kW, 550V, 180 rpm d.c. generator having 92% efficiency. Assume an average flux density in the air gap of about 0.6 Wb/m^2 and ampere conductor per meter to be 35000.
2. Write an algorithm to calculate the following for a 3-phase IM.
 - i) Stator core dimensions
 - ii) Flux/pole
 - iii) No. of stator slots.
3.
 - a) Give the criteria for the selection of stator slots and rotor slots in 3ϕ induction motor.
 - b) Discuss crawling and cogging in three phase induction motor.
4. What do you understand by reliability of a machine? What are different factor affecting reliability? Explain whether these factors can be taken in design of a machine.
5. Differentiate between the Non linear optimization method and linear optimization techniques. How these can be programmed through computer.

OR

Derive the output equation in case of single phase transformer and three phase transformer.

6. Find the main dimension of a 100 MVA, 11 kV, 50 Hz, 150 rpm, three phase water wheel generator. The average gap density of 0.65 Wb/m^2 and ampere conductors per meter are 40,000. The peripheral speed should not exceed 65 m/s at normal running speed in order to limit the run away peripheral speed.

OR

How do you calculate the following for an three phase induction motor?

- i) Area of stator slots
 - ii) Length of mean turn
7. In a 500kW, 440Volt, 375 rpm, 8 pole compound generator, the external diameter of armature is 1.1. mt, gross armature length is 0.3mt, armature slot pitch is 2.5 cm and flux/pole is 0.0875 wb. Determine:
 - i) No. of conductors in armature winding
 - ii) No. of slots in armature
 - iii) Resistance of the armature winding.
- 8 Write short notes on any two of the following.
 - i) Design of the rotor of a turbo alternator.
 - ii) Design of field winding of a D. C machine.
 - iii) Optimal design of induction motor.

PTO